

Fairness and Accuracy Trade-offs in Multiclass Crime Classification

1 Introduction

Machine learning models trained on real-world crime data often learn patterns that reflect existing social and statistical biases. As a result, their predictions may differ systematically across demographic groups. In this work, we focus on gender fairness in a multiclass crime classification task and study how improving fairness affects predictive accuracy. We compare a standard baseline model with a fairness-aware variant, both generated automatically using a prompt-driven AIDE-ML pipeline. This setup allows us to isolate the effect of fairness-oriented design choices while keeping the overall modeling framework unchanged.

2 Dataset

The dataset consists of 6,638 crime records collected from a publicly available Kaggle dataset. The prediction target contains six crime categories: Drug and Weapon Crimes, Miscellaneous, Sexual Crimes, Theft, Vandalism, and Violence. Gender information is provided by the sensitive attribute `Offender_Gender`, with 4,987 male and 1,651 female samples. The data distribution is highly imbalanced, with Violence and Theft accounting for a large majority of cases, which makes both performance evaluation and fairness analysis non-trivial.

3 Model Design

Both models are generated automatically using AIDE-ML through a locally executed runner script. Instead of manually coding the full pipeline, we specify the task using a natural-language prompt that defines data loading, feature preprocessing, model training, and output artifacts. This prompt-driven approach ensures that both models share the same structure and differ only in fairness-related design choices.

3.1 Baseline Model

The baseline model is a standard multiclass Logistic Regression classifier. Numerical features are standardized, and categorical features are one-hot encoded using a unified preprocessing pipeline. The dataset is split into training and test sets using a fixed 80/20 split. This model is optimized solely for predictive performance and does not explicitly account for potential differences between gender groups.

3.2 Fairness-Aware Model

The fairness-aware model uses the same architecture and preprocessing pipeline as the baseline model. To reduce bias caused by strong class imbalance, the classifier is trained with `class_weight="balanced"`. Rather than modifying the feature space, fairness is addressed by encouraging more balanced prediction behavior across gender groups. This design choice allows

us to directly compare how fairness considerations influence model outputs under otherwise identical conditions.

4 Evaluation Metrics

Predictive performance is evaluated using Accuracy and Macro-F1. Accuracy reflects overall correctness, while Macro-F1 assigns equal importance to each crime category and is therefore more informative in the presence of class imbalance.

Fairness is measured using the predicted-rate disparity ratio for each class:

$$\text{Ratio}_c = \frac{\min_g P(\hat{y} = c \mid g)}{\max_g P(\hat{y} = c \mid g)}$$

This metric captures how evenly a model predicts a given crime category across gender groups. Values closer to one indicate more balanced predictions.

5 Results

5.1 Fairness Results

Class	Baseline Ratio	Fair Ratio	Δ Ratio
Miscellaneous	0.47	0.74	+0.27
Theft	0.60	0.43	-0.17
Violence	0.92	0.17	-0.74

Table 1: Gender fairness comparison using predicted-rate disparity ratios.

The fairness-aware model substantially improves gender balance for the *Miscellaneous* category. However, fairness decreases for Theft and Violence, showing that improvements in one class may come at the expense of others. Very rare classes are excluded from fairness analysis due to insufficient sample size.

5.2 Performance Results

Model	Accuracy	Macro-F1
Baseline	0.597	0.219
Fairness-aware	0.316	0.218

Table 2: Overall predictive performance comparison.

While the fairness-aware model shows a large drop in overall accuracy, the Macro-F1 score remains almost unchanged. This indicates that the performance loss mainly affects majority classes, while average performance across all categories is largely preserved.

6 Discussion

The results clearly illustrate the trade-off between fairness and accuracy. Introducing fairness-aware modeling changes how prediction errors are distributed across classes and groups. Rather than uniformly improving all outcomes, fairness gains are achieved by reducing dominance of majority patterns, which leads to lower overall accuracy.

7 Conclusion

This study shows that fairness-aware design in multiclass crime classification can improve gender balance for specific crime categories, but such improvements are not free. Fairness constraints reshape model behavior by redistributing prediction errors, revealing an inherent tension between fairness and accuracy. Future work will focus on more principled, theory-driven prompt design within AIDE-ML to better control this trade-off and improve overall evaluation metrics.